HINT

In different circumstances, you might have to find the 10 °C figure from the graph as well.

c What mass of potassium chloride would crystallise from the solution in **biii** if the temperature fell to 10 °C?

You can use the data in the table to find the solubility at 10 °C. This value is 31.2 g per 100 g of water. That means that 31.2 g of potassium chloride will stay in solution at 10 °C. Since you started with 51.0 g, the rest of it must have formed crystals:

mass of crystals = 51.0 – 31.2 g = 19.8 g

19.8 g of potassium chloride will crystallise out.

END OF CHEMISTRY ONLY

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING (2)

1 What name is given to each of the following changes of state?

a Solid to liquid

b Liquid to solid

c Solid to gas

d Gas to solid

SKILLS INTERPRETATION (5)

2 a Draw diagrams to show the arrangement of the particles in a solid, a liquid and a gas.

b Describe the difference between the movement of particles in a solid and a liquid.

c The change of state from a liquid to a gas can be either evaporation or boiling. Explain the difference between evaporation and boiling.

SKILLS ANALYSIS

3 The questions refer to the substances in the table.

	Melting point/°C	Boiling point/°C
A	–259	–253
B	0	100
C	3700 (sublimes)	
D	–116	34.5
E	801	1413

a Write down the physical states of each compound at

i 30 °C.

ii –100 °C

iii 80 °C

SKILLS PROBLEM SOLVING (6)

b Which substance has the greatest distance between its particles at 25 °C? Explain your answer.

SKILLS REASONING (5)

c Why is no boiling point given for substance **C**?

(6)

d Which liquid substance would evaporate most quickly in the open air at 25 °C? Explain your answer.

SKILLS REASONING

4 Refer to Figure 1.14 on page 7 showing the diffusion experiment.

a Explain why the ring takes a few minutes to form.

b i If you heat a gas, what effect will this have on the movement of the particles?

ii In the light of your answer to i, what difference would you find if you did this experiment outside on a day when the temperature was 2 °C instead of in a warm lab at 25 °C? Explain your answer.

c Explain why the ring was formed nearer the hydrochloric acid end of the tube.

d Suppose you replaced the concentrated hydrochloric acid with concentrated hydrobromic acid. This releases the gas hydrogen bromide (HBr). Hydrogen bromide also reacts with ammonia to form a white ring.

i Suggest a name for the white ring in this case.

ii Hydrogen bromide particles are about twice as heavy as hydrogen chloride particles. What effects do you think this would have on the experiment?

SKILLS CRITICAL THINKING

5 Use the words given below to complete the following paragraph. Each word may be used once, more than once or not at all.

Sodium chloride dissolves in water to form a _____. The water is called the _____ and the sodium chloride is the _____. If the solution is heated to 50 °C some of the water _____ until the solution becomes _____ and sodium chloride crystals start to form.

boils solution solute saturated evaporates solvent condenses

CHEMISTRY ONLY

SKILLS INTERPRETATION

6 The solubility of sodium chlorate in water was measured at a number of different temperatures.

Temperature/°C	0	20	40	60	80	100
Solubility/g per 100 g of water	3	8	14	23	38	55

a Use these figures to plot a solubility curve, with the temperature on the horizontal axis and the solubility on the vertical one.

SKILLS ANALYSIS

b Use your graph to find the solubility of sodium chlorate at 50 °C.

c Determine the maximum mass of sodium chlorate that will dissolve in 40 g of water at 30 °C.

SKILLS PROBLEM SOLVING

d 20 g of sodium chlorate was added to 100 g of water and the mixture heated to about 70 °C. It was then left to cool with the thermometer in the solution. Use your graph to answer the following questions.

i At what temperature would crystals first appear in the solution?

ii If the solution was cooled to 17 °C, calculate the total mass of crystals formed.

END OF CHEMISTRY ONLY